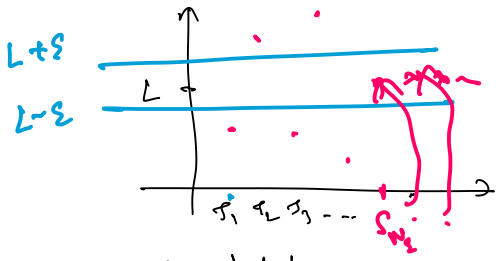
Lecture 21

Wednesday, 23 September 2026 8:23 AM

Recap: $\lim_{n \to \infty} s_n = L$

For ① every $\varepsilon > 0$, $\exists$ a stage $N_\varepsilon \in$ ② $\mathbb{N}$ such that

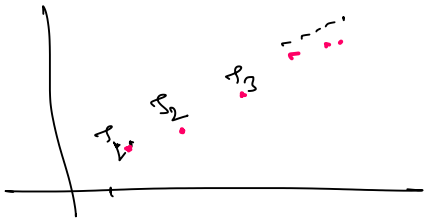
$s_n \in (L-\varepsilon, L+\varepsilon)$ $\forall\, n \geq N_\varepsilon$.



Thm: If $\{s_n\}_{n=1}^{\infty}$ is monotonic and bdd then $\{s_n\}$ is convergent.

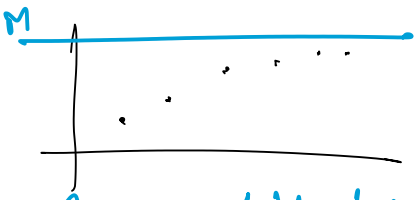
Def: Monotonically ↑

$s_n \leq s_{n+1}$ $\forall\, n \in \mathbb{N}$



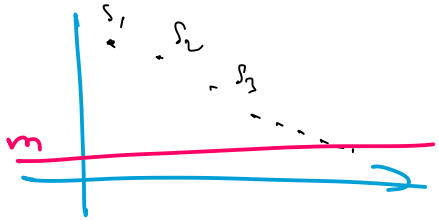
Bdd above:

$s_n \leq M$ $\forall\, n \in \mathbb{N}$



Thm ① If $\{s_n\}$ is monotonically ↑ and bdd above then it is cgt.

Thm ②: If $\{s_n\}$ is monotonically ↓ and bdd below then it is cgt.



(i) $s_n \geq s_{n+1}$ $\forall\, n \in \mathbb{N}$

(ii) $s_n \geq m$ $\forall\, n \in \mathbb{N}$

Eg 1: $s_n = 1 - \frac{1}{n}$ $n \in \mathbb{N}$

$\{1, \frac{1}{2}, \ldots\}$

$= \{0, \frac{1}{2}, \frac{2}{3}, \cdots\}$

Is $\{s_n\}$ increasing?

Consider $f(x) = 1 - \frac{1}{x}$ $\quad x \geq 1$

$f'(x) = \frac{1}{x^2} > 0 \quad \forall x \geq 1$

$\Rightarrow f$ is $\uparrow$.

$\therefore f$ is $\uparrow$ for $x \in \mathbb{N}$

$\Rightarrow s_n = 1 - \frac{1}{n}$ is $\uparrow$

We can see that $s_n = 1 - \frac{1}{n} \leq 1 \quad \forall n \in \mathbb{N}$

$\Rightarrow \{s_n\}$ is monotonically $\uparrow$ and bdd above

$\therefore$ By the thm, $\{s_n\}$ cgt.

Eg2): $s_n = e^{-n}$

$s_{n+1} \leq s_n$? $\geq$

$\frac{s_{n+1}}{s_n} = \frac{e^{-(n+1)}}{e^{-n}} = e^{-1} = \frac{1}{e} \leq 1$

$\Rightarrow s_{n+1} \leq s_n \quad \forall n \in \mathbb{N}$

$\Rightarrow \{s_n\}$ is monotonically $\downarrow$.

Also, $s_n = e^{-n} \geq 0 \quad \forall n \in \mathbb{N}$

$\Rightarrow \{s_n\}$ is bdd below

Thm $\Rightarrow \{s_n\}$ cgt.

Eg3): $s_n = \frac{n}{2^n}$ $\quad (n \geq 1)$

$\frac{s_{n+1}}{s_n} = \frac{n+1}{2n} = \frac{1}{2} + \frac{1}{2n} \leq 1$

$\Rightarrow s_{n+1} \le s_n \quad \forall n \ge 1$

$\Rightarrow \{s_n\}$ is monotonically $\downarrow$

$s_n \ge 0 \quad \forall n \in \mathbb{N}$

Thm $\Rightarrow \{s_n\}$ cgs.

Eg: $s_n = n^{1/n}$

$\lim_{n \to \infty} s_n = ?$

$f(x) = x^{1/x} \quad \forall x \ge 1$

$f(x) = e^{\frac{1}{x} \ln x}$

$f'(x) = e^{\frac{\ln x}{x}} \left(\frac{1 - \ln x}{x^2} \right)$

$\ge 0 \quad \le 0 \quad \forall x \ge e$

≤ 0

$\Rightarrow f$ is $\downarrow$ for $x \ge 3$.

$\Rightarrow \{s_n\}$ is decreasing

& $s_n \ge 0 \quad \forall n \in \mathbb{N}$

Thm $\Rightarrow \{s_n\}$ cgs.

Check

$\lim_{x \to \infty} x^{1/x} = 1$

Eg: $s_n = \sqrt{n+1} - \sqrt{n}$

$\{s_n\}$ is $\downarrow$

& $s_n \ge \boxed{0} \quad \forall n \in \mathbb{N}$

$\Rightarrow \{s_n\}$ cgs.

Ex: $s_n = \left(1 + \frac{a}{n}\right)^n \qquad \lim_{n \to \infty} s_n = e^a$?

